

2024 Information Retrieval and Extraction HW2

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Questions

1. (10%) Describe your approach to **data processing and information retrieval**. Please choose at least 2 of any IR methods and compare their performance.

Data processing

Because the original data was very messy, containing many irrelevant symbols such as HTML, punctuation, or special characters, I used a script to remove HTML, convert to lowercase, remove punctuation and special characters, and remove stopwords. Then, I saved the cleaned data for further use.

Information retrieval

We applied three different Information Retrieval (IR) methods to find the most relevant evidence for each claim: **TF-IDF**, **BM25**, and **BERTScore**.

TF-IDF

Since the dataset did not provide evidence sentences, I was unable to train a new TF-IDF model. Therefore, I used the `TfidfVectorizer` from the `sklearn` library for implementation. TF-IDF first converts the preprocessed sentences and the preprocessed claim into vectors, then calculates the cosine similarity between them. Finally, the top 10 most similar and unique sentences are selected as the evidence sentences.

```
if IR == "TF-IDF":
    # Create TF-IDF representation
    vectorizer = TfidfVectorizer(stop_words='english')
    # Fit the vectorizer on the sentences and transform them
    tfidf_matrix = vectorizer.fit_transform(preprocessed_sentences)
    # Transform the claim into the TF-IDF space
    claim_vector = vectorizer.transform([preprocessed_claim])
    # Calculate cosine similarity between the claim and all sentences
    cosine_similarities = cosine_similarity(claim_vector, tfidf_matrix).flatten()
    # Get the top most relevant sentences
    top_indices = np.argsort(cosine_similarities)[::-1][:TOP_K]
```

BM25

Since the dataset did not provide evidence sentences, I was unable to train a new BM25 model. Therefore, I used BM25Okapi from the rank_bm25 library for implementation. First, I used the tokenized sentences to build a BM25 model, then calculated the BM25 scores between the tokenized claim and these sentences. Finally, the top 10 highest-scoring and unique sentences were selected as the evidence sentences.

```
elif IR == "BM25":
    # Tokenize sentences for BM25
    tokenized_sentences = [sentence.split() for sentence in preprocessed_sentences]
    # Initialize BM25 model
    bm25 = BM25Okapi(tokenized_sentences)
    # Tokenize the claim
    tokenized_claim = preprocessed_claim.split()
    # Get BM25 scores for all sentences
    bm25_scores = bm25.get_scores(tokenized_claim)
    # Get the top most relevant sentences
    top_indices = np.argsort(bm25_scores)[::-1][:TOP_K]
```

BERTScore

Since the dataset did not provide evidence sentences, I was unable to train a new BERTScore model. Therefore, I used the score function from the bert_score library for implementation. First, I used the score function to calculate the precision, recall, and F1 score between the preprocessed claim and all preprocessed sentences. Then, I selected the top 10 highest F1-scoring and unique sentences as the evidence sentences.

```
elif IR == "BERTScore":
    # Calculate BERTScore between the claim and all sentences
    P, R, F1 = score(preprocessed_sentences, [preprocessed_claim] * len(preprocessed_sentences), lang='en',
    # Get the top 5 most relevant sentences based on BERTScore F1
    f1_scores = F1.tolist()
    top_indices = sorted(range(len(f1_scores)), key=lambda i: f1_scores[i], reverse=True)[:TOP_K]
```

Compare their performance

Since the dataset did not provide evidence sentences, I could not directly measure which IR method was better. Therefore, I compared their performance by training classification models using the evidence sentences retrieved by each IR method.

The experimental results showed that TF-IDF (bert-base-uncased_1) performed the best, followed by BM25 (bert-base-uncased_2), and finally BERTScore (bert-base-uncased_5).

TF-IDF outperformed BM25 and BERTScore due to its effective lexical matching in a dataset where precise word overlap was more beneficial than length normalization or semantic similarity.

	bert-base-uncased_5.csv Complete · 4d ago · bert-base-uncased_5	0.52421	☐
	bert-base-uncased_4.csv Complete · 4d ago · bert-base-uncased_4	0.52836	☐
	bert-base-uncased_3.csv Complete · 4d ago · bert-base-uncased_3	0.53572	☐
	bert-base-uncased_2.csv Complete · 4d ago · bert-base-uncased_2	0.54385	☐
	bert-base-uncased_1.csv Complete · 4d ago · bert-base-uncased_1	0.54605	☐

2. (10%) Describe your approach to **claim prediction**. Details such as model selection, hyperparameters should be provided. If you use **LLMs**, try using techniques such as in-context learning, chain-of-thought prompting or any techniques that can improve the quality of generation and compare their performance.

Fine-tuned model

Model selection

For claim prediction, I used **bert-base-uncased**, **bert-large-uncased**, **roberta-base**, and **bart-base** as pre-trained models, which are well-suited for sequence classification tasks like fact-checking. The ability of these models to understand contextual relationships between words made them ideal for analyzing claims along with their corresponding evidence sentences. The classification head of each model was adapted to predict three classes: "False" (0), "Partial True" (1), and "True" (2).

Training process

To train the model, the claim and the retrieved evidence sentences (from the IR methods) were concatenated and fed into BERT. Specifically, the claim and evidence were separated by a [SEP] token to maintain clarity. Each input sequence was then tokenized with a maximum length of 512 tokens, truncating longer sequences.

Hyperparameters

I trained the model using the default hyperparameters provided by the transformers Trainer:

- Batch Size: 8
- Learning Rate: 2e-5
- Number of Epochs: 3

- Warmup Steps: 500
- Weight Decay: 0.01
- Optimizer: AdamW

Compare their performance

In the end, only bert-base-uncased was successfully trained, while the other models failed for unknown reasons.

	bert-large-uncased_1.csv Complete · 3d ago · bert-large-uncased_1	0.20228	☐
	roberta-base_1.csv Complete · 4d ago · roberta-base_1	0.20228	☐

LLM

I used the OpenAI API with GPT-4o-mini and designed four types of in-context learning: zero-shot, one-shot, few-shot, and chain-of-thought:

Zero-shot

I did not provide any reference examples to the LLM. Instead, I directly gave it the claim and evidence sentences to predict the label. The prompt was as follows:

```
The following is a fact-checking task. You will be given a claim followed by evidence sentences.
Your task is to classify the claim as one of the following: 0 (False), 1 (Partial True), or 2 (True).
Only output 0, 1, or 2!

Claim: {claim}

Evidence Sentences:
{evidence_sentences}

Label:
```

One-shot

In addition to the claim and evidence sentences, I provided one reference example. The prompt was as follows:

```
The following is a fact-checking task. You will be given a claim followed by evidence sentences.
Your task is to classify the claim as one of the following: 0 (False), 1 (Partial True), or 2 (True).
Only output 0, 1, or 2!
```

Here is an example you can refer to:

Example 1

Claim: Video shows that "Joe Biden participated in 'Black Face' skit."

Evidence Sentences:

joe biden
joe biden introduced black singer michael jackson democratic dinner
1985 joe biden refers black band member michael jackson
1980s fundraising dinner joe biden singled black vocalist
biden
user clip 1985 joe biden refers black band member michael jackson
clips live shows
upcoming live shows
people video
joe biden jr us senator delaware

Label: 0

Test

Claim: {claim}

Evidence Sentences:
{evidence_sentences}

Label:

Few-shot

In addition to the claim and evidence sentences, I provided three reference examples. The prompt was as follows:

```
The following is a fact-checking task. You will be given a claim followed by evidence sentences.
Your task is to classify the claim as one of the following: 0 (False), 1 (Partial True), or 2 (True).
Only output 0, 1, or 2!
```

Here are three examples you can refer to:

Example 1

Claim: "Video shows that "Joe Biden participated in 'Black Face' skit.""

Evidence Sentences:

"joe biden"
"joe biden introduced black singer michael jackson democratic dinner"
"1985 joe biden refers black band member michael jackson"
"1980s fundraising dinner joe biden singled black vocalist"
"biden"
"user clip 1985 joe biden refers black band member michael jackson"
"clips live shows"
"upcoming live shows"
"people video"
"joe biden jr us senator delaware"

Label: 0

Example 2

Claim: "While Act 10 allowed (Milwaukee County) to save some money, it was millions short of what we expected."

Evidence Sentences:

"senate allowed act"
"county would get hit reduction 28 million state"
"hole left fill roughly half budget parks"

Chain-of-thought

In addition to the claim, evidence sentences, and three reference examples, I also provided the reasoning behind the classification. The prompt was as follows:

```
The following is a fact-checking task. You will be given a claim followed by evidence sentences.
Your task is to classify the claim as one of the following: 0 (False), 1 (Partial True), or 2 (True).
Let's think step by step! Please tell me the reason first, and then tell me the label of the classifi

Here are three examples you can refer to:

# Example 1

Claim: "Video shows that "Joe Biden participated in 'Black Face' skit.""

Evidence Sentences:
"joe biden"
"joe biden introduced black singer michael jackson democratic dinner"
"1985 joe biden refers black band member michael jackson"
"1980s fundraising dinner joe biden singled black vocalist"
"biden"
"user clip 1985 joe biden refers black band member michael jackson"
"clips live shows"
"upcoming live shows"
"people video"
"joe biden jr us senator delaware"

Reason:
The evidence sentences focus on Joe Biden's actions, such as introducing a black singer (Michael Jack

Label: 0
```

Compare their performance

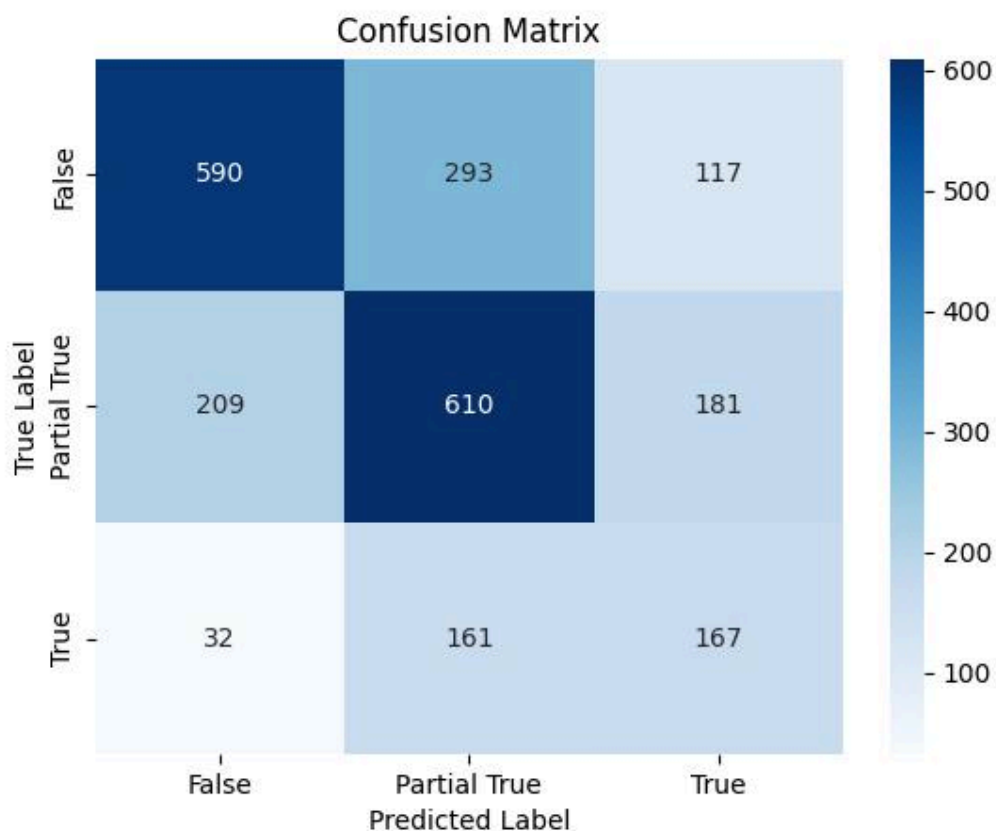
From the experiment, I observed that one-shot performed the best, followed by zero-shot, few-shot, and chain-of-thought. This result was different from what I had expected, indicating that a more complex prompt does not necessarily lead to better performance. Moreover, the performance of the LLM was lower than that of the fine-tuned model, showing that relying solely on LLM + in-context learning is not sufficient to solve this problem effectively.

	gpt-4o-mini_chain-of-thought.csv Complete · 2d ago · gpt-4o-mini_chain-of-thought	0.40615	☐
	gpt-4o-mini_few-shot.csv Complete · 3d ago · gpt-4o-mini_few-shot	0.43054	☐
	gpt-4o-mini_one-shot.csv Complete · 3d ago · gpt-4o-mini_one-shot	0.45314	☐
	gpt-4o-mini_zero-shot.csv Complete · 3d ago · gpt-4o-mini_zero-shot	0.43442	☐

3. (10%) Do **error analysis** or **case study**. Is there anything worth mentioning while checking the mispredicted data? Share with us. Anytime you try to make a conclusion about the data or model, you should provide concrete data example.

Error analysis

For error analysis, I used **TF-IDF** along with **bert-base-uncased** to predict the labels on the validation dataset, and then plotted the confusion matrix of the true and predicted labels, as shown in the figure below.



From the confusion matrix, we can observe that the proportion of instances where the label is **True** but the prediction is **Partial True** is the highest (almost 45%). I suspect this is because the number of instances labeled as True is much smaller compared to Partial True and False (only 360 instances for True, whereas Partial True and False each have 1,000). This imbalance in the training dataset led to poorer performance on the True label. Upsampling or downsampling might help address this issue.

Case study

```
"metadata": {
  "claimant": "Hillary Clinton",
  "claim": "\"The top 25 hedge fund managers (are) making more than all of America's kindergarten teachers combined.\"\"",
  "id": 26992,
  "premise_articles": {
    "https://time.com/3920332/transcript-full-text-hillary-clinton-campaign-launch/": "26992 1.json",
    "http://www.institutionalinvestorsalpha.com/Article/3450284/The-2015-Rich-List-The-Highest-Earning-Hedge-Fund-Managers-are-Making-More-Than-All-of-Americas-Kindergarten-Teachers-Combined-": "26992 2.json",
    "http://www.bls.gov/oes/current/oes252012.htm#nat": "26992 3.json"
  },
  "top_relevant_sentences": [
    "top 25 hedge fund managers making americas",
    "prosperity cant ceos hedge fund managers",
    "kindergarten teachers combined often paying lower tax rate",
    "252012 kindergarten teachers except special education",
    "students teachers",
    "fireside chat better alternative hedge fund replication",
    "teachers desk",
    "teachers 252050 substitute teachers shortterm 253031",
    "1 year 25",
    "valueacts new ceo mason morfit shows hedge fund succession work"
  ]
},
"label": {
  "rating": 2,
  "original_rating": "true",
  "id": 26992
}
```

This is an example where the label is **True** but the prediction is **Partial True**. I found that the evidence sentences in this example do not directly support the claim ("The top 25 hedge fund managers are making more than all of America's kindergarten teachers combined"). The evidence sentences do not provide actual data to verify this statement. I suspect that the prediction error occurred due to poor quality of the retrieved evidence sentences. To solve this issue, a stronger retriever is needed to find better evidence.

```
"metadata": {
  "claimant": null,
  "claim": "A photograph shared widely on social media in July 2019 shows former U.S. Secretary of Labor Alex Acosta and his family.",
  "id": 26984,
  "premise_articles": {
    "https://t.co/0tDuijwnpl": "26984 1.json",
    "https://twitter.com/eehouls/status/1149008324198719489?ref_src=twsrc%5Etfw": "26984 2.json"
  },
  "top_relevant_sentences": [
    "link alternate pt",
    "link alternate fr",
    "link alternate de",
    "link alternate nl",
    "link alternate id",
    "see list supported browsers help center",
    "refresh httpstwittercomeehoulsstatus1149008324198719489photo1",
    "link alternate ru",
    "try",
    "link alternate ko"
  ]
},
"label": {
  "rating": 2,
  "original_rating": "true",
  "id": 26984
}
```

This is another example where the label is **True** and the prediction is also **True**. Interestingly, the evidence sentences are of poor quality, but the prediction was still correct. This suggests that the retriever's poor performance may have led the model to learn incorrect information, highlighting the need for improvement in the retriever's capabilities.